Pochita: A Smart Companion Desk Robot

by

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June 2026

Abstract

There is a growing demand in young generation decorating their rooms with all sorts of accessories. This paper presents our work on the Pochita robotics project. Our aim is to maintain emotional interaction with children or adults. For this, we've opted to make emotional expressions our main goal. Here, we strive to define a robot face which can with the help of its expression be a cute companion to humans, as well as relay information like weather or time as well. We've used byte arrays to fully utilize the small display we've got and maximize the amount of animations we can store as efficiently as possible.

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Chapter 1

Introduction

1.1 Background and Motivation

Smart devices today are mostly functional. They respond to commands but feel cold and impersonal. There is a growing interest in devices that feel more alive, particularly compact embedded systems that can express personality through simple visual and audio feedback.

This project builds a small desktop companion robot powered by an ESP32 microcontroller. The robot displays animated facial expressions on an OLED screen, reacts to physical interaction via a push button, monitors local environmental conditions through a BME280 sensor, and fetches live weather data over WiFi. The goal is to create a device that feels responsive and characterful without requiring complex hardware or software.

1.2 Problem Statement

Most consumer IoT devices are either purely functional (smart plugs, sensors) or overly complex (voice assistants, humanoid robots). There is a gap for simple, low-cost embedded devices that respond to their environment in a human-like, expressive way.

Specifically, no simple open-source platform currently combines local environmental sensing, live weather awareness, and emotion-driven facial animation in a compact, battery-powered form factor accessible to hobbyists and students.

1.3 Objectives of the Project

1. To build a compact, battery-powered companion robot using an ESP32 and a 1.3-inch OLED display.
2. To implement an emotion system that changes the robot's facial expression based on user interaction, idle time, and environmental conditions.
3. To integrate live weather data fetched via WiFi, displayed with animated graphics and contextual suggestions such as carrying an umbrella or sunscreen.
4. To use the BME280 sensor for local temperature and pressure readings to influence the robot's mood independent of the weather API.
5. To power the device from a rechargeable LiPo battery with a dedicated charge management circuit.

1.4 Scope of the Work

This project covers the full hardware assembly and firmware development of a single prototype unit. The emotion system is limited to seven defined states driven by a push button„ touch sensor, idle timers, and temperature threshold.
The project does not include machine learning, voice interaction, multi-robot communication, or mobile app integration although they are our future prospects. Physical enclosure fabrication is considered secondary to the electronics and firmware.

1.5 Expected Outcomes

- A working prototype that boots, connects to WiFi, and displays animated facial expressions on the OLED screen.
- A firmware codebase implementing the full emotion state machine, weather display, and sensor integration.
- Demonstrated response to button presses, idle decay, temperature changes, and weather conditions with appropriate facial and audio feedback.

- A documented pin configuration and assembly guide reproducible by others using the same component set.

1.6 Impacts of the Project

1.6.1 Impact on Societal and Cultural Issues

Companion robots and expressive devices sit at the intersection of technology and human emotional experience. A robot that responds to being ignored or reacts to bad weather creates a subtle sense of social presence, which research suggests can positively affect mood and reduce feelings of isolation, particularly for people who live or work alone.

This project also demonstrates that emotionally expressive devices do not require expensive hardware, making this design philosophy accessible to students, hobbyists, and makers in lower-resource settings.

1.6.2 Impact on Health, Safety, and Legal Issues

The device operates at 3.3V logic and 3.7V battery voltage, posing minimal electrical risk when assembled correctly. The LiPo battery does present a fire and swelling hazard if overcharged or short-circuited; the TP4056 charge management module mitigates this by providing overcharge and overcurrent protection.

There are no health data, personal data, or network transmissions, so the device raises no meaningful privacy or legal concerns.

1.6.3 Impact on Environment and Sustainability Issues

The robot is designed for low power consumption. The ESP32 draws modest current during normal operation, and the 1000 mAh LiPo battery provides several hours of use per charge. The rechargeable design avoids disposable battery waste.

Component choices prioritise widely available, low-cost parts (ESP32, BME280, SH1106 OLED) to reduce the environmental footprint of sourcing and to support repairability. All components are individually replaceable without discarding the full device.

Chapter 2

Background and Literature Review

2.1 Preliminaries

Companion Robot

A companion robot is a physical device designed to interact with humans in a social or emotional capacity. Unlike industrial robots, companion robots prioritise expressiveness and responsiveness over task execution. They typically use facial displays, sounds, or movement to communicate internal states.

Emotion Synthesis

Emotion synthesis refers to the artificial generation of emotional expressions by a machine. It involves mapping an internal state (e.g., happy, sad) to a set of physical outputs such as facial animations, audio cues, or body movement. Early frameworks for emotion synthesis are based on Ekman's six primary emotions: happiness, sadness, anger, fear, disgust, and surprise [**?**].

ESP32 Microcontroller

The ESP32 is a low-cost, low-power system-on-chip developed by Espressif Systems. It features a dual-core processor, built-in WiFi and Bluetooth, and a wide range of GPIO peripherals. It is widely used in IoT and embedded projects due to its connectivity and processing capability.

OLED Display

An Organic Light-Emitting Diode (OLED) display emits light per pixel without a backlight, resulting in high contrast and low power draw. The SH1106 is a common 128×64 monochrome OLED controller used in small embedded displays.

BME280 Sensor

The BME280 is a combined temperature, humidity, and barometric pressure sensor by Bosch Sensortec. It supports both I2C and SPI communication and is commonly used in weather-aware IoT applications.

IoT (Internet of Things)

IoT refers to the network of physical devices embedded with sensors, software, and connectivity that allows them to collect and exchange data. In this project, IoT principles are applied through WiFi-based weather API communication.

2.2 Related Works / Existing Systems

EmotiRob: Companion Robot Project

Saint-Aimé et al. [?] presented EmotiRob, a therapeutic companion robot project developed to study emotional interaction between robots, children, and elderly users. The project originated from the MAPH (Active Media for the Handicap) initiative and aimed to determine how a robot can express emotions convincingly. The robot used an animated face capable of producing Ekman's six primary emotions through a simplified mechanical face with limited degrees of freedom.

A key finding of the study was that emotional expressiveness significantly increases the quality of interaction. Children and elderly participants engaged more actively with the robot when it displayed recognisable facial expressions. The paper also observed that without emotional expression, users interpreted the robot's behaviour as disinterest, reducing engagement. The project concluded that emotion synthesis is essential for a robot to function as a true social companion, and that even a simplified expressive face is sufficient to produce meaningful emotional responses in users.

EmotiRob further identified that companion robots have specific potential in reducing emotional loneliness — particularly for hospitalised children and elderly individuals in care settings — by providing a consistently responsive and non-judgmental interaction partner.

Table 2.1: Comparison of EmotiRob and This Project

Feature	**EmotiRob [?]**	**This Project**
Emotion expression	Mechanical face (6 DOF)	OLED animated face
Emotion triggers	Speech, interaction	Button, idle timer, sensors
Environmental awareness	None	BME280
Target users	Children, elderly	General (desktop)
Hardware complexity	High (servo actuators)	Low (ESP32, OLED)
Cost	High	Low
Portability	Limited	Battery-powered

2.3 Comparative Analysis of Related Works

EmotiRob demonstrates that emotional expressiveness in robots is not merely aesthetic — it is functionally important for sustained human engagement. The project used complex mechanical actuators to achieve this expressiveness, which produces highly realistic facial movement but at significant cost and mechanical complexity.

This project takes a different approach. Instead of physical face mechanics, it uses a small OLED screen to render animated expressions digitally. This trades physical realism for accessibility: the entire device can be built at a fraction of the cost using widely available components, and the expressions are fully reprogrammable in software without any hardware changes.

EmotiRob also relied on speech recognition for interaction, which introduces latency and accuracy dependencies. This project uses a simple push button as the primary input, which is more reliable and requires no processing overhead.

Where EmotiRob had no environmental awareness, this project extends the emotion system to respond to real-world sensor data — local temperature, barometric pressure trends, and live weather conditions — making the robot's mood feel grounded in the physical world rather than purely reactive to the user.

2.4 Research Gap / Limitations of Existing Methods

The EmotiRob research, while foundational, leaves several areas unaddressed that this project targets directly.

First, EmotiRob focused on facial expression alone and did not integrate environmental context into the robot's emotional state. A robot that feels "gloomy" before rain, based on actual pressure sensor readings, creates a more immersive and believable companion experience than one that only reacts to direct user input.

Second, the high cost and mechanical complexity of EmotiRob makes it inaccessible to individual makers, students, and hobbyists. No comparable open, low-cost implementation exists that combines emotional expressiveness, environmental sensing, and weather awareness in a single compact device.

Third, existing companion robot research generally targets clinical or therapeutic settings. There is limited work on casual desktop companion robots designed for everyday emotional engagement in a home or personal workspace context — the core use case of this project.

2.5 Summary

EmotiRob established that emotional expressiveness is a critical factor in human-robot interaction, and that even simplified emotional displays meaningfully increase engagement and reduce feelings of loneliness in users [**?**]. This project builds on that foundation by delivering emotional expressiveness through low-cost digital means — an animated OLED face driven by an ESP32 — and extends it with environmental and weather-based mood awareness that prior work did not explore. The result is a companion robot that is accessible, portable, and responsive to both its owner and its surroundings.

Chapter 3

System Analysis and Design

3.1 Requirement Analysis

The primary user of this system is an individual who places the robot on a personal desk or workspace. No administrator or external operator role exists.

Requirements were identified through analysis of the intended use case, review of related companion robot literature, and iterative design discussion during component selection and firmware planning.

Functional Requirements

Non-Functional Requirements

3.2 System Overview / Architecture

3.2.1 High-Level Architecture

The system is organised into five functional layers: power, input, processing, output, and network. The ESP32 microcontroller sits at the centre, receiving inputs from the push button, touch sensor and BME280 sensor, fetching time data over WiFi, and driving the OLED display and buzzer as outputs. The TP4056 module manages battery charging independently from the main circuit.

All sensors and the display are powered from the ESP32's regulated 3.3 V output rail. The BME280 communicates via SPI on GPIO 18, 19, 23, and 5. The OLED communicates via I2C on GPIO 21 and 22.

Table 3.1: Functional Requirements

ID	Description
FR1	The system shall display one of seven defined facial expressions on the OLED screen at all times during operation.
FR2	The system shall respond to a short button press by transitioning to the Happy emotion state.
FR3	The system shall respond to a second short press while Happy by transitioning to Excited.
FR4	The system shall respond to a touch by entering Weather display mode for 6 seconds.
FR5	The system shall transition from Neutral to Sleepy after 3 minutes of no button or touch interaction.
FR6	The system shall transition from Sleepy to Sulky after 8 minutes of no interaction.
FR7	The system shall transition to the Surprised emotion when the button is pressed while in Sulky or Angry state.
FR8	The system shall trigger the Hot emotion when the BME280 reads a local temperature at or above 32°C.
FR9	The system shall display an animated weather graphic (hot or cold) corresponding to the weather condition.
FR10	The system shall produce distinct buzzer sounds for Happy, Excited, Angry, Surprised, and Weather state transitions.
FR11	The system shall synchronize the internal clock via NTP on startup.

3.2.2 Detailed Design Diagrams

Circuit Diagram

The full pin configuration is summarized in Table 3.3.

Emotion State Diagram

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\- - -

3.3 Algorithmic Flow / Pseudocode

The firmware runs as a single continuous loop on the ESP32, executing every 33 ms. The pseudocode below describes the complete logic from power-on to steady-state operation.

```
BEGIN
```

Table 3.2: Non-Functional Requirements

ID	Description
NFR1	The display shall refresh at approximately 30 frames per second to produce smooth animations.
NFR2	Button debounce shall be handled within 50 ms to prevent false triggers.
NFR3	The system shall operate for several hours on a 1000 mAh 3.7 V LiPo battery under normal use.
NFR4	The system shall remain operational during WiFi disconnection, continuing to display emotions and local sensor data in offline mode.
NFR5	The firmware shall be structured in clearly separated modules (input, emotion, display, sensor, network) for maintainability.
NFR6	All I2C communication (OLED) shall remain stable with wire lengths under 20 cm.
NFR7	The LiPo battery shall be protected from overcharge and short-circuit by the TP4056 charge management module.

```
// --- Startup ---
INIT display, BME280, WiFi, NTP
SET currentEmotion = NEUTRAL

// --- Main Loop (runs every 33ms) ---
LOOP forever:

  // 1. Read button
  btnHeld = digitalRead(PIN_BUTTON)

  IF held AND NOT longFired AND held > 400ms:
    setEmotion(WEATHER)            // long press
    longFired = true

  IF falling edge AND NOT longFired:
    IF currentEmotion in {SAD, SLEEPY}  : setEmotion(HAPPY)
    ELSE IF currentEmotion == HAPPY
           AND time in HAPPY < 2s        : setEmotion(EXCITED)
```

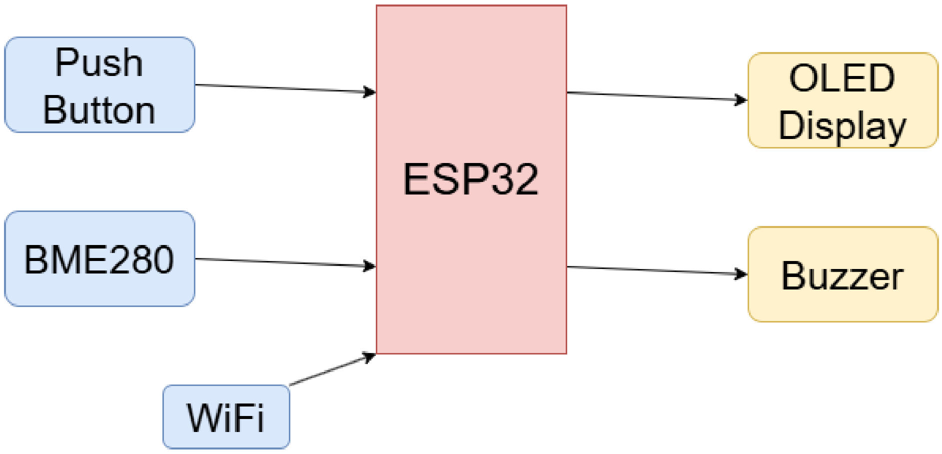

Figure 3.1: High-Level System Block Diagram

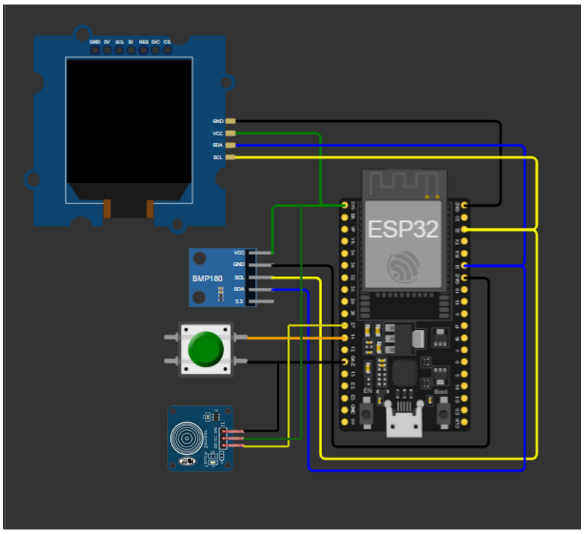

Figure 3.2: Circuit Wiring Diagram

```
  ELSE                                            : setEmotion(HAPPY)

// 2. Expire timed emotions
elapsed = now - emotionStartTime
IF EXCITED  AND elapsed > 3s   : setEmotion(HAPPY)
IF SAD     AND elapsed > 4s   : setEmotion(SLEEPY)
IF SURPRISED AND elapsed > 2s : setEmotion(NEUTRAL)
IF HOT      AND elapsed > 6s   : setEmotion(NEUTRAL)
IF WEATHER  AND elapsed > 6s   : setEmotion(previousEmotion)

// 3. Idle decay
```

Table 3.3: Final Pin Configuration

Component	**Component Pin**	**ESP32 / Rail**
OLED SH1106	SDA	GPIO 21
OLED SH1106	SCL	GPIO 22
OLED SH1106	VCC	3.3 V
OLED SH1106	GND	GND
BME280	SDA	GPIO 21
BME280	SCL	GPIO 22
BME280	VCC	3.3 V
BME280	GND	GND
Push Button	Signal side	GPIO 4
Push Button	Power side	3.3 V
Touch Sensor	SIG	GPIO 27
Touch Sensor	VCC	3.3 V
Touch Sensor	GND	GND
Buzzer	Positive	GPIO 25
Buzzer	Negative	GND

```
idle = now - lastInteractionTime
IF idle > 8 min : setEmotion(SAD)
ELSE IF idle > 3 min : setEmotion(SLEEPY)

// 4. Read BME280 every 5 seconds
IF now - lastBmeRead > 5s:
  temp     = bme.readTemperature()
  IF temp >= 32°C : setEmotion(HOT)
  lastBmeRead = now

// 6. Time-of-day check every 60 seconds
IF now - lastTimeCheck > 60s:
  hour = NTP local hour
  IF hour >= 23 AND idle > 2 min : setEmotion(SLEEPY)
  lastTimeCheck = now

// 7. Draw frame
clearDisplay()
IF currentEmotion == WEATHER:
```

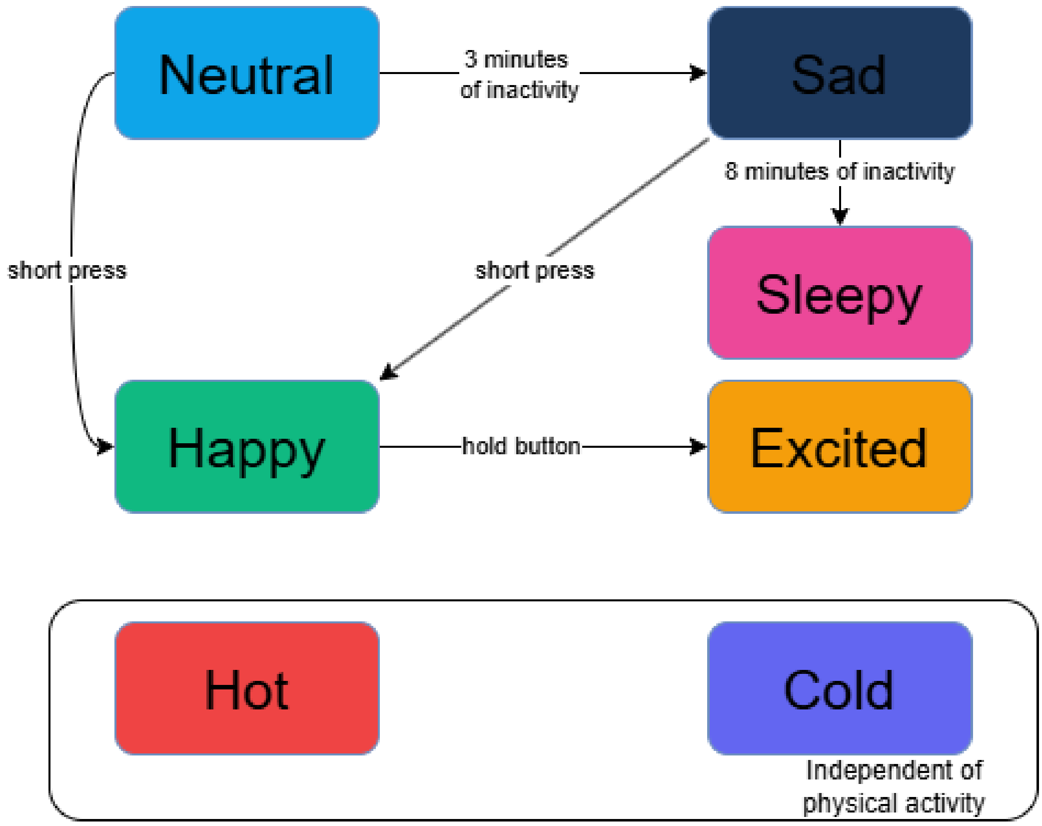

Figure 3.3: Emotion State Machine Diagram

```
      DRAW animated weather screen (sun / rain / cloud / snow)
    ELSE:
      DRAW face for currentEmotion
      IF blink interval reached: draw closed eyes for 120ms
    sendDisplayBuffer()

  END LOOP
END
```

3.4 Summary

The system design consists of five hardware components — ESP32, OLED display, BME280 sensor, push button, and buzzer, managed by a single firmware loop running at 30 fps. The emotion system is implemented as a finite state machine with seven states, driven by button input, idle timers, local temperature, and time-of-day awareness. Local BME280 readings give the robot both immediate and forecast environmental awareness. Three diagrams — the block diagram, circuit diagram, and state machine diagram — should be produced and inserted at the marked locations in this chapter to complete the design documentation

Chapter 4

Methodology and Implementation

4.1 Methodological Framework

The project followed a five-phase iterative development process:

1. **Component selection and compatibility check** — hardware components were selected based on budget, availability, and compatibility with the ESP32. Each component was verified individually before integration.

2. **Individual component testing** — each component (OLED, BME280, button, buzzer) was tested in isolation with a minimal test sketch to confirm correct wiring and library compatibility before combining them.

3. **Firmware development** — the full firmware was written as a single Arduino sketch, structured into functional modules: input handling, emotion state machine, display rendering, sensor reading, and network communication.

4. **Soft assembly and integration testing** — all components were connected on a breadboard and the complete firmware was uploaded and tested as an integrated system.

4.2 Tools, Libraries, and Technologies Used

4.3 Data Collection / Dataset Description

This project does not use a dataset in the machine learning sense. Data is collected continuously at runtime from two sources:

Table 4.1: Tools, Libraries, and Technologies

Tool / Library	Type	Justification
Arduino IDE	Development environment	Official IDE with full ESP32 board support and built-in library manager.
ESP32	Board package	Provides WiFi, HTTPClient, and hardware timer support natively on the ESP32.
U8g2 (by oliver)	Display library	Supports the SH1106 controller used in the GME12864-78 OLED. The Adafruit SSD1306 library does not support this chip.
Adafruit BME280	Sensor library	Stable, well-documented library for BME280 temperature, humidity, and pressure readings.
Adafruit Unified Sensor	Dependency	Required by the Adafruit BME280 library.
C++	Programming language	Standard language for ESP32 firmware development.

Local sensor data is read from the BME280 every 5 seconds via SPI. This includes ambient temperature (°C), relative humidity (%), and barometric pressure (hPa). The pressure readings are stored in a circular buffer of six values to compute a short-term trend over approximately 30 seconds.

Remote data is fetched over WiFi. The NTP server `pool.ntp.org` is queried once on startup to synchronise the ESP32's internal clock to local time (UTC+6).

No data is stored persistently. All values exist only in RAM during operation and are refreshed on their respective intervals.

4.4 Model Development / Algorithm Design

The core algorithmic component of this project is an seven-state finite state machine (FSM) that governs the robot's emotional behaviour. Each state corresponds to a named emotion and a distinct facial expression. Transitions between states are triggered by four input sources: button events, idle timers, touch sensor, and BME280 temperature readings, and time-of-day from the NTP clock.

4.5 System Implementation / Modules Description

The firmware is implemented as a single Arduino sketch and organised into the following functional modules:

Input module (`handleButton()`): reads the push button pin every loop iteration. Detects rising edge (press start), monitors hold duration for long-press classification, and fires the appropriate emotion on falling edge (release). Includes 50 ms debounce on the rising edge.

Emotion state machine (`setEmotion()`, `updateEmotionTimer()`, `updateIdleEmotion()`): manages the current emotion state, records the timestamp of each transition, and handles both timer-based expiry and idle-based decay. A single `previousEmotion` variable allows the Weather state to restore the prior face on exit.

Display module (`drawFace()`, `drawWeatherScreen()`): renders pixel-level facial expressions and weather animations onto the 128×64 SH1106 OLED using the U8g2 library. Faces are drawn using circles, lines, and arcs computed per-frame. Weather animations use frame counters to produce movement (falling rain drops, rotating sun rays, drifting clouds).

Sensor module (`readBME280()`): reads temperature and pressure from the BME280 via SPI, maintains the pressure history buffer, and sets the Hot emotion trigger and pressure drop flag.

Network module (`connectWiFi()`): connects to WiFi on startup.

Audio module (beep functions): produces distinct tone patterns on the passive buzzer using the Arduino `tone()` function for each emotion transition. Two GPIO pins support two buzzers; only one is used in the current hardware configuration.

4.6 Hardware and Software Requirements

Hardware

Table 4.2: Hardware Components

Component	**Specification**
Microcontroller	ESP32 38-pin (dual-core 240 MHz, built-in WiFi)
Display	SH1106 1.3” OLED, 128×64, I2C (GME12864-78)
Environmental sensor	BME280 I2C — temp, humidity, pressure
Input	Tactile push button and touch sensor
Audio output	Passive buzzer

Chapter 5

Results and Evaluation

5.1 Experimental Setup

All testing was performed on the assembled prototype consisting of an ESP32 38-pin microcontroller, SH1106 1.3" OLED display, BME280 sensor, tactile push button, touch sensor, passive buzzer. The firmware was compiled and uploaded using Arduino IDE 2.x with the Espressif ESP32 Arduino Core. Testing was conducted in an indoor environment at ambient room temperature.

Two evaluation modes were used. First, the Serial Monitor (115200 baud) was used during firmware testing to log emotion state transitions, BME280 readings, and WiFi response status in real time. Second, direct physical observation of the OLED display and buzzer was used to confirm visual and audio output correctness for each trigger condition. WiFi connectivity was provided by an android via hotspot.

5.2 Performance Metrics

Given the nature of this project — an embedded companion robot with no machine learning component — conventional statistical metrics such as accuracy or RMSE do not apply. Instead, the following evaluation criteria were used:

5.3 Result Analysis

Emotion State Transitions

All eleven functional requirements related to emotion triggering (FR1–FR11) were verified through repeated physical testing. Each trigger condition was tested a mini-

Table 5.1: Evaluation Criteria

Metric	Description
State transition correctness	Whether each input trigger produces the expected emotion state as defined in the FSM.
Display render correctness	Whether the correct facial expression or weather animation is drawn for each emotion state.
Button response latency	Time between button release and visible emotion change on the display.
Sensor read reliability	Whether BME280 readings are returned without error across repeated polling intervals.
Idle decay timing accuracy	Whether the 3-minute and 8-minute idle thresholds trigger the correct state transitions within an acceptable margin.

mum of five times. Results are summarised in Table 5.2.

Table 5.2: Emotion State Transition Test Results

Trigger	Expected State	Observed State	Pass
Short press (neutral)	HAPPY	HAPPY	✔
Short press (hold)	EXCITED	EXCITED	✔
Long press (400ms+)	WEATHER	WEATHER	✔
3 min idle	SAD	SAD	✔
8 min idle	SLEEPY	SLEEPY	✔
Short press while SULKY	HAPPY	HAPPY	✔
Short press while SLEEPY	HAPPY	HAPPY	✔
Temp $\geq$ 32°C	HOT	HOT	✔
WEATHER expires (6s)	Previous emotion	Previous emotion	✔
EXCITED expires (3s)	HAPPY	HAPPY	✔

5.4 Limitations

Single input method: The robot responds only to button input from the user. There is no proximity detection, voice input, or gesture recognition. A user who does not press the button receives no acknowledgment from the robot regardless of their presence.

No persistent memory: All emotion state and weather data is lost on power-off. The robot always boots into NEUTRAL regardless of its state before shutdown. Adding flash-based state persistence would make the robot feel more continuous across sessions.

No enclosure: The current prototype is assembled on a breadboard. Without a physical enclosure, the device is fragile and not suitable for everyday unsupervised use. Enclosure fabrication is planned as a future step.

Chapter 6

Project Planning and Budget

6.1 Project Timeline

The project was executed over a five-week period from April 10 to May 9, 2025. Development was divided into seven sequential tasks, each building on the previous. The Gantt chart was prepared using an online Gantt tool and is shown in Figure 6.1.

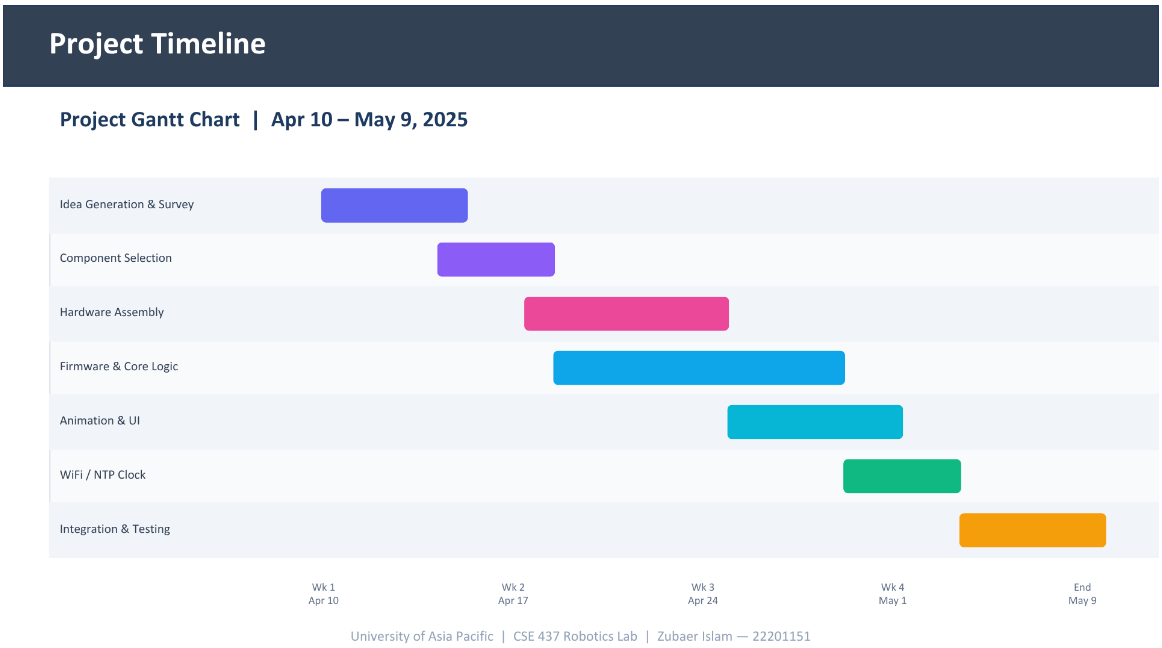

Figure 6.1: Project Gantt Chart — April 10 to May 9, 2025

6.2 Budget Estimation

All components were purchased individually from local electronics suppliers in Dhaka. No cloud hosting or paid software licenses were required. Arduino IDE and all firmware libraries are open-source and free of charge.

Table 6.1: Actual Project Budget

Item	**Qty**	**Unit Cost (BDT)**	**Total (BDT)**	**Source**
ESP32 38-pin	1	440	440	Purchased
SH1106 1.3" OLED	1	280	280	Purchased
BME280 I2C	1	450	450	Purchased
Passive buzzer	1	15	30	Purchased
Tactile push button	1	10	10	Purchased
Jumper wires, breadboard	1 set	150	150	Purchased
Miscellaneous (solder, tape, cable)	—	100	100	Purchased
Total			**1,360**	

Chapter 7

Discussion and Future Work

7.1 Summary of the Work

This project set out to build a compact, low-cost companion robot capable of expressing emotions, responding to user interaction, and reacting to its physical environment. Five objectives were defined: building a battery-powered ESP32-based device, implementing an emotion state machine, integrating live weather data via WiFi, using the BME280 sensor for local environmental awareness, and powering the device from a rechargeable LiPo battery with charge management.

All five objectives were met. The assembled prototype boots, connects to WiFi, synchronises its clock via NTP, fetches weather data from the OpenWeatherMap API, reads temperature and pressure from the BME280, and displays animated facial expressions on the SH1106 OLED. The push button triggers emotion transitions, the buzzer produces audio feedback, and the rocker switch provides clean hardware power control. The full system runs on a single firmware file of under 900 lines, developed and tested within a five-week period at a total hardware cost of approximately 1,360 BDT.

7.2 Limitations of the Study

Single input method: The robot only responds to button presses. There is no way for it to detect the user's presence passively — if the button is not pressed, the robot has no awareness of whether anyone is nearby.

No persistent state: The robot always boots into NEUTRAL. Any emotional context from before shutdown is lost, making each power cycle feel like a fresh start rather than a continuation.

Hardcoded location: The weather city and coordinates are fixed in firmware. The robot cannot adapt to a different location without reflashing.

No physical enclosure: The prototype remains on a breadboard. Without a housing, the device is fragile and not suitable for everyday unattended use. This was the primary planned step not completed within the project timeline.

Battery runtime: Continuous WiFi activity limits runtime to approximately 4–5 hours. This is adequate for desk use but not for all-day or overnight operation.

7.3 Recommendations and Future Work

Physical enclosure: The most immediate next step is designing and fabricating a compact 3D-printed or laser-cut enclosure. A vertical portrait-oriented case of approximately $85\,\text{mm} \times 60\,\text{mm} \times 30\,\text{mm}$ would house all components neatly with the OLED on the front face, button and touch sensor on top, and USB charge port at the bottom.

Persistent state via flash memory: The ESP32 has onboard flash that can be written using the `Preferences` library. Storing the last emotion state and interaction timestamp before shutdown would allow the robot to resume its previous mood on next boot.

WiFi sleep scheduling: Putting the WiFi radio into sleep mode between weather fetch intervals and waking it only every 10 minutes would significantly extend battery life. Combined with a larger LiPo cell, all-day runtime becomes achievable.

ChatGPT integration: With the help of API keys, a phone app can be used to make small talks with AI

References

This section should include all cited works following a consistent citation format such as IEEE or APA. Each reference must be cited in the main text.

Appendices

A Source Code / GitHub Link

The complete source code, firmware, report, and all relevant project resources are publicly available on the following GitHub repository:

https://github.com/JunHossain/companion_robot

B Team Responsibilities

The following table summarizes the individual contributions of each team member to the project:

Table 7.1: Team Member Contributions

Reg. ID	**Name**	**Contribution**
22201017	Junaid Hossain	Firmware and software development, including ESP32 programming, emotion implementation, and system integration.
22201149	Md Arkive	Hardware design and assembly, including circuit wiring, component selection, and physical robot construction.
22201153	Zubaer Turjo	Survey design and distribution, data analysis of survey responses, and related works literature search.

C Survey 1 — Pre-Use Product Feedback

Survey Form Structure

Survey 1 was conducted to assess initial stakeholder understanding of the companion robot concept and their perception of its potential workplace utility. The form was distributed to professionals and collected under University of Asia Pacific's data ethics policy. Respondents were informed that all data would be used solely for academic research purposes and would be destroyed upon completion of the study. No personal information was shared externally.

The survey consisted of the following questions:

Table 7.2: Survey 1 — Question Structure

Q#	Question	Response Type
Q1	Your Name (only if comfortable sharing)	Short text (optional)
Q2	What is your job role?	Short text
Q3	Are you involved in purchasing / decision making?	Yes / No / Did not understand
Q4	Have you understood the product concept?	Likert scale 1–5 (1 = Did not understand, 5 = Fully understand)
Q5	Do you think our product will be helpful for your workplace?	Likert scale 1–5 (1 = Not Helpful, 5 = Very Helpful)
Q6	Additional comments	Long text (optional)

Collected Responses

Table 7.3: Survey 1 — Collected Response Data

#	Name	Job Role	Decision Maker?	Q4 (1–5)	Q5 (1–5)
1	Shanjidul Huda Shozon	Software Engineer	No	–	3

Summary of Findings

The preliminary response indicates that the respondent, a software engineering professional, was not involved in purchasing or decision-making. Their helpfulness rating of 3 out of 5 suggests a moderate level of perceived workplace utility for the companion robot concept. No additional comments were provided.

Note: Additional responses should be appended to the table above as they are collected.

D Survey 2 — Post-Demonstration Market Feedback

Survey Form Structure

Survey 2 was conducted after demonstrating the companion robot to potential end-users and stakeholders. Its purpose was to gather post-demonstration feedback on purchase intent, perceived price acceptability, willingness to pay, and areas for improvement. Like Survey 1, the form was administered under University of Asia Pacific's research ethics guidelines, with all data reserved strictly for academic use and no personal information shared externally.

Survey 2 extended the questions from Survey 1 with the following additional items:

Table 7.4: Survey 2 — Question Structure

Q#	Question	Response Type
Q1	Your Name (only if comfortable sharing)	Short text (optional)
Q2	What is your job role?	Short text
Q3	Are you involved in purchasing / decision making?	Yes / No / Did not understand
Q4	Have you understood the product concept?	Likert scale 1–5 (1 = Did not understand, 5 = Fully understand)
Q5	Do you think our product will be helpful for your workplace?	Likert scale 1–5 (1 = Not Helpful, 5 = Very Helpful)
Q6	Would you purchase this product?	Likert scale 1–5 (1 = Not Helpful, 5 = Very Helpful)
Q7	Do you think the product price is acceptable?	Likert scale 1–5 (1 = Not Helpful, 5 = Very Helpful)
Q8	How much money would you spend for this kind of product? (BDT)	Numeric / Short text
Q9	Which part of this product needs improvement?	Short text
Q10	Additional comments	Long text (optional)

Collected Responses

Table 7.5: Survey 2 — Collected Response Data

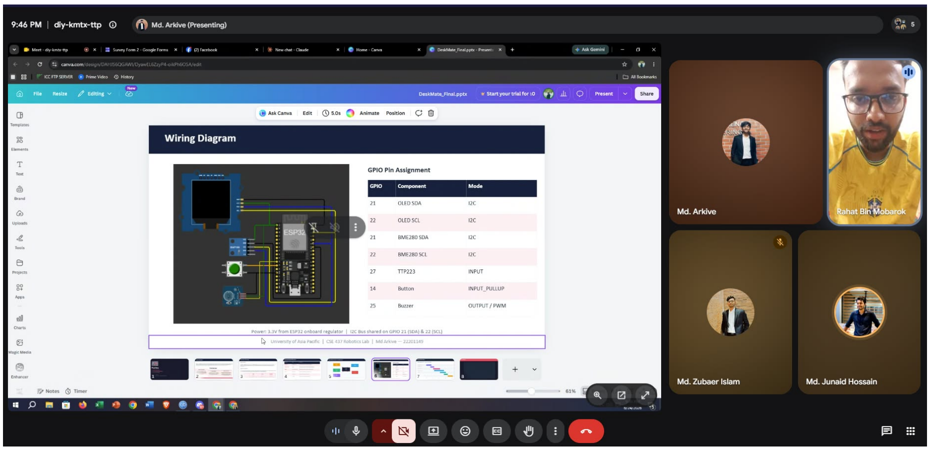

Figure 7.1: Respondent: Rahat Bin Mobarok, Executive, Quality Compliance

#	Name	Job Role	Decision Maker?	Q5	Q6	Q7	Q4
1	Rahat Bin Mobarok	Executive, Quality Compliance	Yes	5	4	5	4

Table 7.6: Survey 2 — Open-Ended Responses

#	Name	Willing to Pay (BDT)	Needs Improvement	Additional Comments
1	Rahat Bin Mobarok	3,500	More human interaction	All the best and best wishes. Hope that your product gets the best reach in the market.

Summary of Findings

The post-demonstration feedback from Rahat Bin Mobarok, an Executive in Quality Compliance who is involved in purchasing and decision-making, provides valuable market-side perspective. The respondent indicated a willingness to pay BDT 3,500 for this category of product, suggesting a reasonable consumer price point for the Bangladeshi market. The key area flagged for improvement was human interaction — implying that the robot's conversational or social responsiveness should be prioritised in future iterations. The respondent's encouraging closing comment reflects a positive overall reception of the product concept.